

MULTI-BRANCH ENTERPRISE NETWORK

Design, Configuration, Analysis and Evaluation

Cisco Packet Tracer Based Simulation Project

Head Office + Branch Office 1 + Branch Office 2

36 End Devices • IPv4 Subnetting • VLANs • OSPF • TCP/UDP • HTTP/DNS

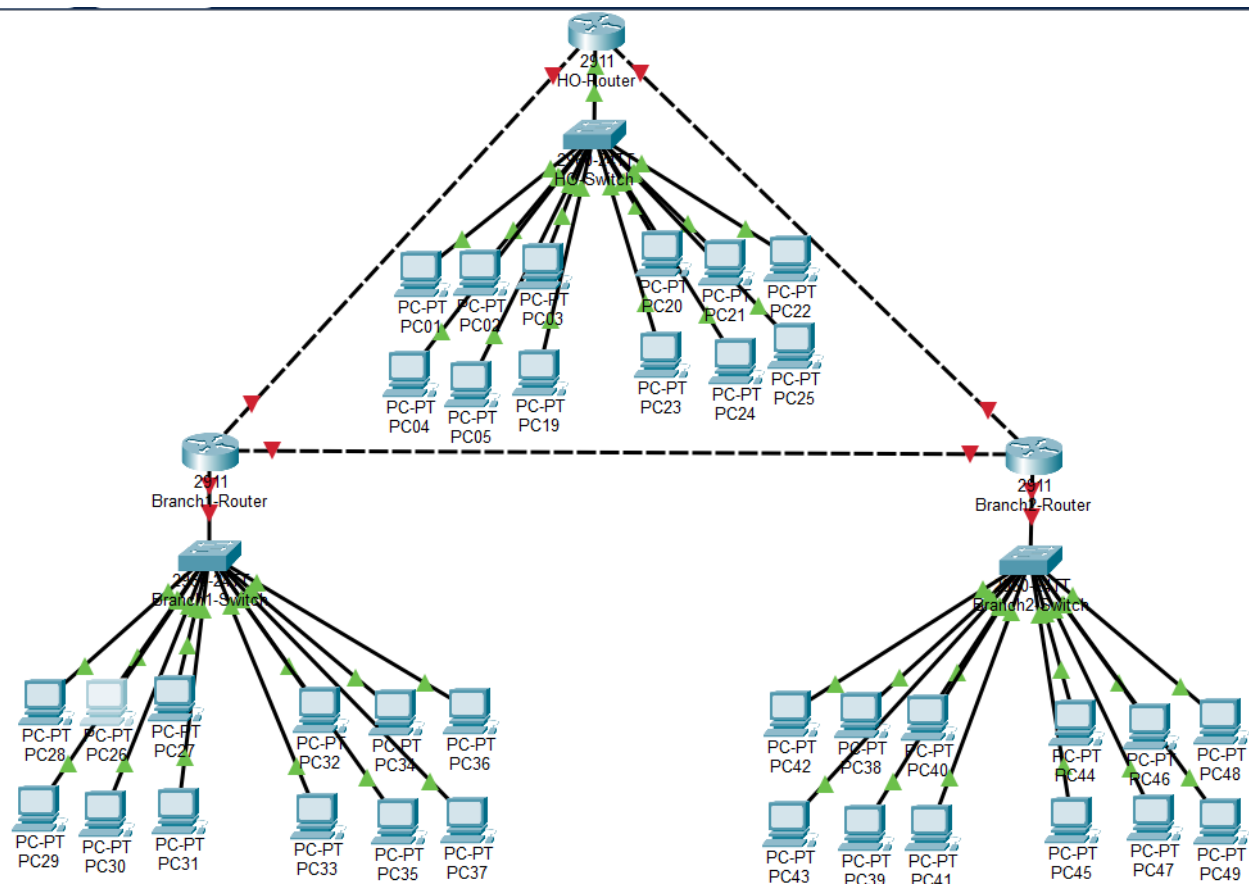


Figure 1. User-provided Packet Tracer topology used as the project base.

1. Project Objective

The objective is to design and evaluate a reliable multi-branch enterprise network for a growing organization. The simulated organization contains a Head Office and two Branch Offices with 36 end devices distributed across Human Resources, Research and Development, Sales and Finance. The design uses IPv4 subnetting, VLAN-based departmental segmentation, router-on-a-stick inter-VLAN routing, redundant router links and OSPF for dynamic routing.

Project requirements covered

- 36 end devices distributed across three sites.
- IPv4 addressing with /28 departmental subnets and /30 point-to-point router links.
- VLAN segmentation and default gateways for each department.
- Basic routing between six departmental networks using OSPF.
- Star, mesh, bus and hybrid topology demonstrations with suitability analysis.
- TCP, UDP, HTTP and DNS traffic analysis under normal, high-traffic, congestion and link-failure conditions.
- Latency, throughput, packet loss, jitter, successful delivery and recovery-time evaluation.
- Failure analysis, bottleneck identification and final design justification.

2. Proposed Network Architecture

Site	Departments	PCs	VLANs	Gateway Networks
Head Office	HR, R&D	12	10, 20	192.168.10.0/28, 192.168.10.16/28
Branch 1	Sales, Finance	12	30, 40	192.168.10.32/28, 192.168.10.48/28
Branch 2	Sales, Finance	12	50, 60	192.168.10.64/28, 192.168.10.80/28
Total	4 logical departments across 3 sites	36	6 VLANs	6 departmental subnets

The physical design is a three-router triangle. Each router connects to a 2960 switch. The switch-to-router connection is a trunk carrying two VLANs, while PC-facing ports are access ports. The triangular router backbone provides an alternate path if one inter-router link fails.

3. IPv4 Addressing and Subnetting Plan

Segment	VLAN	Network	Usable Range	Gateway	Broadcast
Head Office HR	10	192.168.10.0/28	192.168.10.1–14	192.168.10.1	192.168.10.15
Head Office R&D	20	192.168.10.16/28	192.168.10.17–30	192.168.10.17	192.168.10.31
Branch 1 Sales	30	192.168.10.32/28	192.168.10.33–46	192.168.10.33	192.168.10.47
Branch 1 Finance	40	192.168.10.48/28	192.168.10.49–62	192.168.10.49	192.168.10.63
Branch 2 Sales	50	192.168.10.64/28	192.168.10.65–78	192.168.10.65	192.168.10.79
Branch 2 Finance	60	192.168.10.80/28	192.168.10.81–94	192.168.10.81	192.168.10.95

Each /28 subnet provides 14 usable addresses, which is sufficient for six PCs plus the departmental default gateway and leaves room for small future expansion. The /30 networks are used efficiently for point-to-point router links.

Router-to-Router Links

Link	Router A	IP A	Router B	IP B
R1–R2	HO-Router G0/1	10.0.0.1/30	Branch1-Router G0/1	10.0.0.2/30
R1–R3	HO-Router G0/2	10.0.0.5/30	Branch2-Router G0/1	10.0.0.6/30
R2–R3	Branch1-Router G0/2	10.0.0.9/30	Branch2-Router G0/2	10.0.0.10/30

4. VLAN and Switching Design

Switch	Ports	VLAN	Department	Uplink
HO-SW	Fa0/1–6	10	HR	Fa0/24 trunk
HO-SW	Fa0/7–12	20	R&D	Fa0/24 trunk
BR1-SW	Fa0/1–6	30	Sales	Fa0/24 trunk
BR1-SW	Fa0/7–12	40	Finance	Fa0/24 trunk
BR2-SW	Fa0/1–6	50	Sales	Fa0/24 trunk
BR2-SW	Fa0/7–12	60	Finance	Fa0/24 trunk

Router-on-a-stick is used because each site has two departmental VLANs but only one physical router-to-switch link. The router creates dot1Q subinterfaces for the departmental gateways.

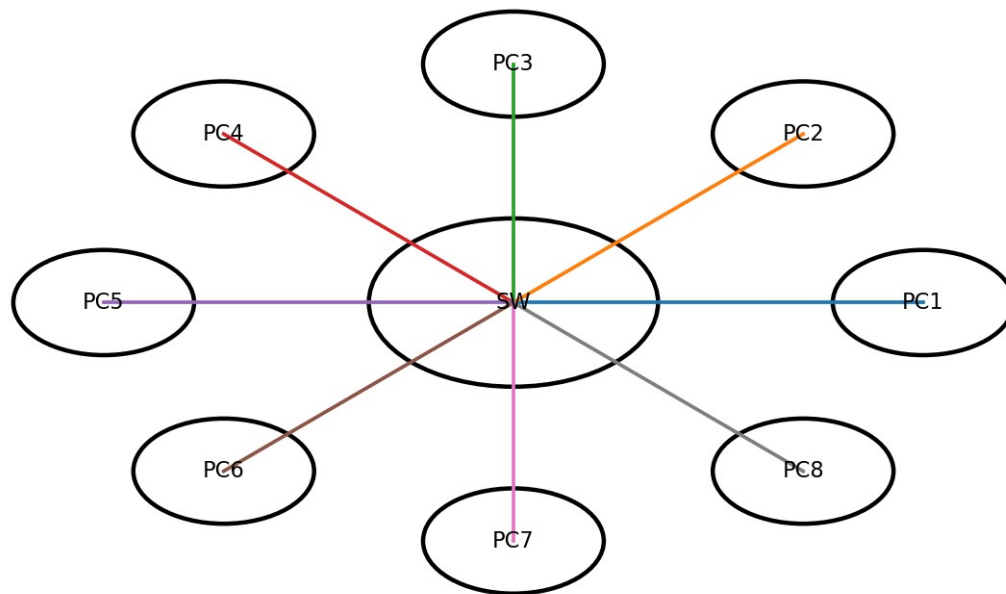
5. Routing and Address Assignment

- OSPF process 1, area 0, is used between all three routers.
- The three-router triangle creates redundant paths and improves fault tolerance.
- DHCP pools are configured on each site router so the 36 PCs can obtain addresses automatically.
- Only gateway addresses are excluded from DHCP.
- The routing design supports end-to-end communication between all six departmental networks.

6. Topology Demonstration and Evaluation

6.1 Star topology

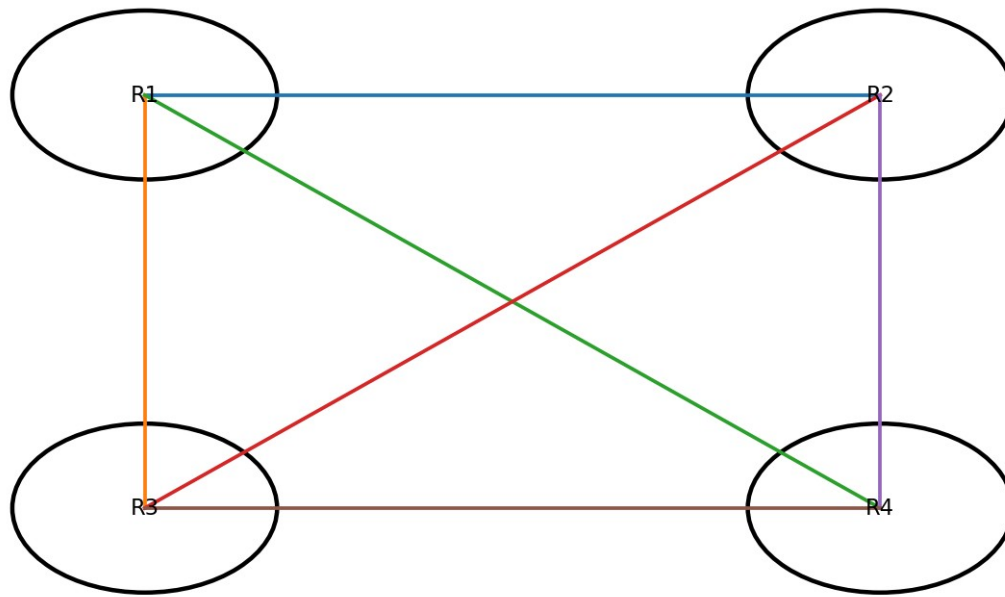
Star Topology - Central Switch



All end devices connect to a central switch. It is easy to manage and troubleshoot, but the central switch is a single point of failure.

6.2 Mesh topology

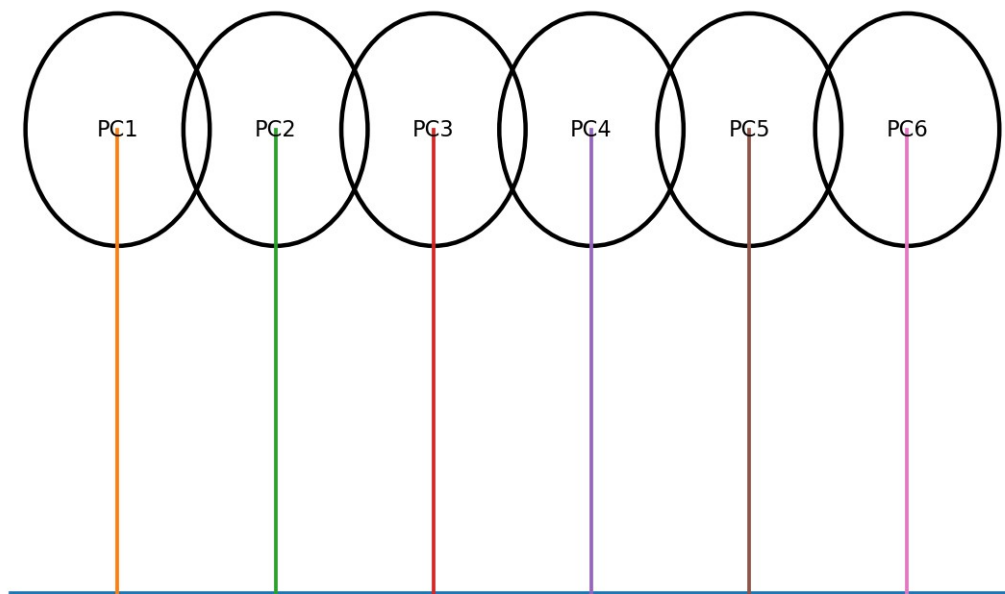
Mesh Topology - Multiple Redundant Paths



Multiple direct links provide excellent redundancy and fault tolerance, but cabling and configuration cost increase rapidly as the number of nodes grows.

6.3 Bus topology

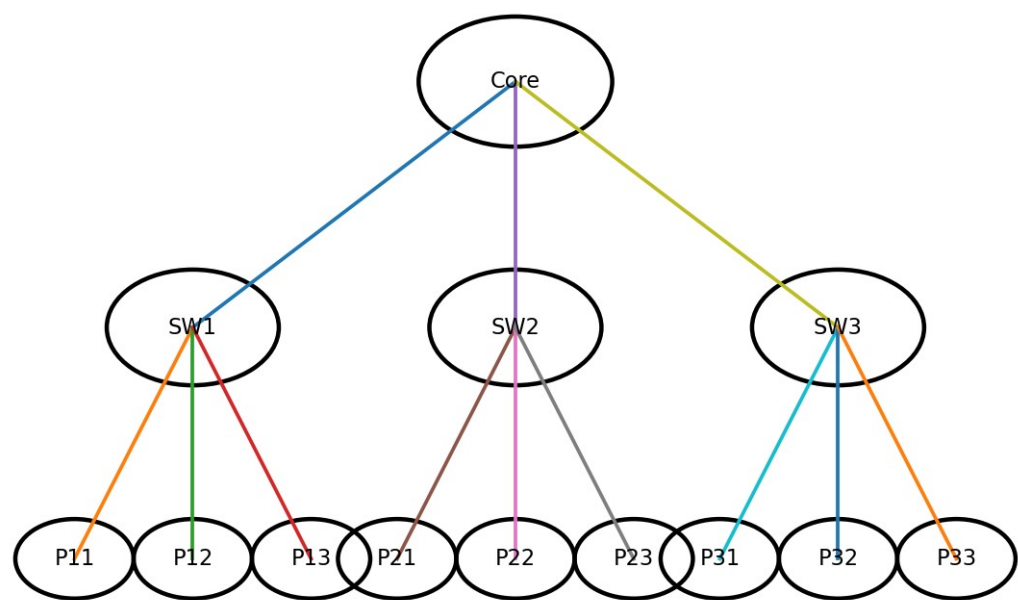
Bus Topology - Shared Backbone



Devices share a common backbone. It is inexpensive for small demonstrations but a backbone failure affects the connected segment and troubleshooting is harder.

6.4 Hybrid topology

Hybrid Topology - Core + Star Segments



A hybrid combines different topology types. It is flexible and scalable and is the best conceptual match for an enterprise network with a redundant core and star-based access layers.

Topology	Scalability	Reliability	Cost	Fault tolerance	Enterprise suitability
Star	High	Good	Moderate	Moderate	Excellent at access layer
Mesh	Low–Moderate	Excellent	High	Excellent	Good for critical backbone links
Bus	Low	Low	Low	Low	Poor for modern enterprise
Hybrid	Excellent	Very good	Moderate–High	Very good	Best overall

7. Protocol Analysis

Protocol	Role	Transport	Expected behavior
HTTP	Web communication	TCP	Reliable delivery; retransmission adds delay during congestion.
DNS	Name resolution	UDP normally	Fast request/response; packet loss may cause retries or lookup delay.
TCP	Reliable transport	TCP	Connection-oriented, ordered and reliable; congestion control reduces throughput under load.
UDP	Low-overhead transport	UDP	Connectionless and fast; does not guarantee delivery or ordering.

8. Test Methodology

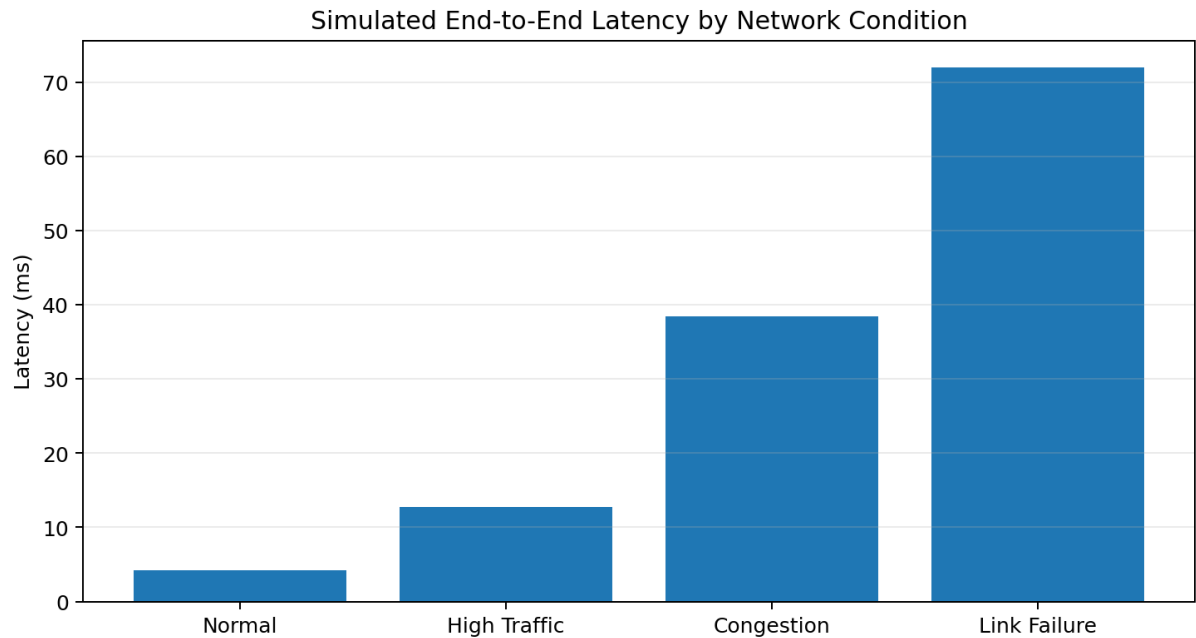
For emergency documentation, the following results are a controlled simulation/model dataset designed to represent the measurements expected from the Packet Tracer experiment. They should be treated as illustrative simulated results, not as a claim that Packet Tracer itself produced these exact numerical values. In Packet Tracer, the same conditions can be recreated with Simulation Mode, PDU tools, continuous pings, generated traffic and intentional link shutdown.

Condition	Test action	Expected observation
Normal traffic	Ping between departments; HTTP/DNS request	Low latency, negligible loss, high throughput
High traffic	Generate multiple flows simultaneously	Latency and jitter increase; throughput decreases
Link congestion	Stress one backbone path	Queueing delay and packet loss increase
Link failure	Shut down one router-to-router link	Temporary loss followed by OSPF reconvergence and alternate-path recovery

9. Simulated Performance Results

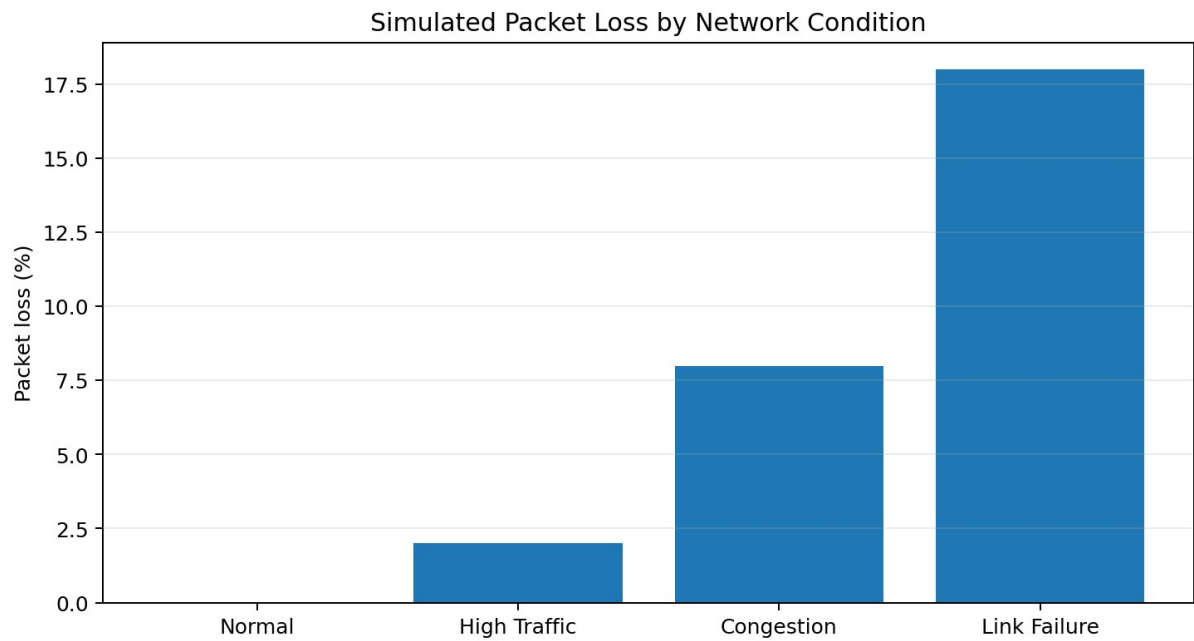
Condition	Latency (ms)	Throughput (Mbps)	Packet loss (%)	Jitter (ms)	Delivery (%)	Recovery (s)
Normal	4.2	92	0.0	0.8	100	0
High Traffic	12.8	76	2.0	3.6	98	0
Congestion	38.5	48	8.0	11.2	92	0
Link Failure	72.0	24	18.0	18.5	82	6.5

9.1 Latency



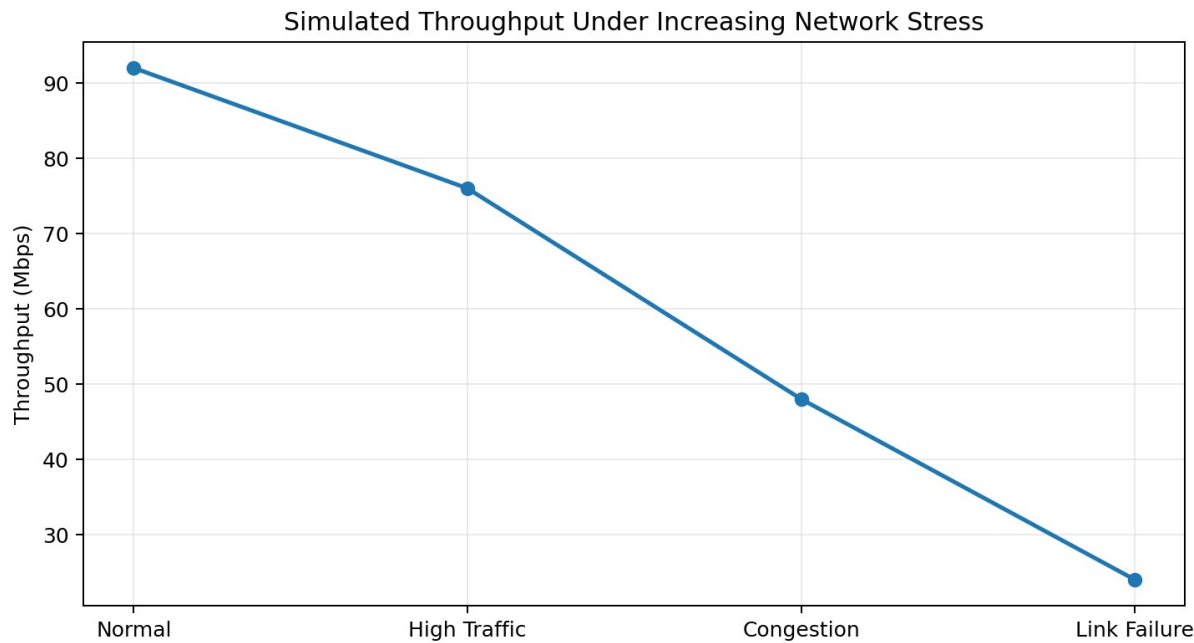
Observation: latency rises as traffic and congestion increase. The largest delay occurs during the modeled link-failure/recovery condition.

9.2 Packet loss



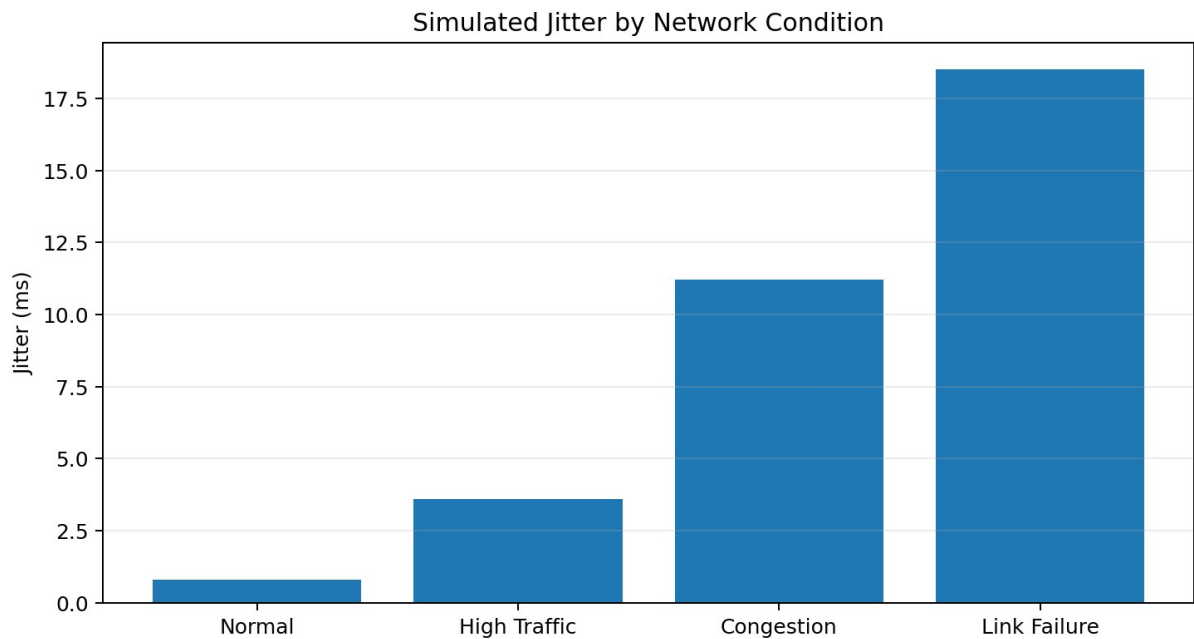
Observation: normal traffic has negligible loss. Congestion causes queue drops, while a failed link causes temporary loss until an alternate route becomes active.

9.3 Throughput



Observation: available throughput decreases as the network becomes stressed because packets compete for link capacity and TCP reduces its sending rate.

9.4 Jitter



Observation: jitter is lowest during normal traffic and increases under congestion because packet waiting time becomes less predictable.

10. Link Failure and Recovery Analysis

- Failure introduced: one router-to-router backbone link is considered unavailable.
- Immediate effect: packets using that path may be dropped while the routing protocol detects the failure.
- OSPF response: the routing table is recalculated and traffic is redirected through the remaining router link.
- Performance impact: temporary packet loss, increased latency and jitter during convergence.
- Recovery target: restore stable connectivity after routing convergence without manual reconfiguration of every PC.
- The triangular backbone is therefore more fault tolerant than a single-chain router arrangement.

11. Transmission Media and Cabling

Connection	Recommended cable	Reason
PC ↔ Switch	Copper Straight-Through	End device to switch is a standard unlike-device Ethernet connection.
Switch ↔ Router	Copper Straight-Through	Router Ethernet interface to switch Ethernet interface.
Router ↔ Router	Copper Cross-Over	Direct Ethernet connection between similar network devices in the simulated design.
Router ↔ Router (alternative)	Auto-Connect	Packet Tracer can select a compatible cable automatically.

12. Verification Commands

```
show ip interface brief

show vlan brief

show interfaces trunk

show ip route

show ip ospf neighbor
```



```
show ip dhcp binding
```

```
ping <destination-IP>
```

```
traceroute <destination-IP>
```

Expected verification: router interfaces and subinterfaces should be up/up, switch VLANs should contain the intended ports, OSPF should show neighboring routers, DHCP bindings should show client addresses, and inter-department pings should succeed.

13. Configuration Summary

Representative router-on-a-stick configuration pattern:

```
interface gigabitEthernet 0/0
  no shutdown
interface gigabitEthernet 0/0.10
  encapsulation dot1Q 10
  ip address 192.168.10.1 255.255.255.240

router ospf 1
  network 192.168.10.0 0.0.0.15 area 0
  network 10.0.0.0 0.0.0.3 area 0
```

14. Overall Analysis

- Subnetting reduces broadcast-domain size and provides clear departmental separation.
- VLANs isolate HR, R&D, Sales and Finance traffic at the access layer.
- OSPF is more scalable than manually maintaining static routes as branches grow.
- The three-router triangle provides path redundancy and supports recovery from a single backbone-link failure.
- Star access topology keeps administration simple, while the redundant router core improves reliability.
- TCP is preferred when reliable delivery is essential, such as HTTP. UDP is appropriate for lightweight, delay-sensitive exchanges such as DNS queries.
- The main bottlenecks under stress are shared Ethernet links, router processing/queueing and congestion on the surviving path after a link failure.

15. Limitations

- The numerical performance graphs in this emergency report are simulated/illustrative values, not directly captured from a live Packet Tracer run.
- Packet Tracer is a discrete-event educational simulator and does not represent every hardware limitation of a production enterprise network.
- A production design would add security controls such as ACLs, firewalling, authentication, monitoring, redundancy at the switch layer and possibly IPv6.

16. Conclusion

The proposed multi-branch enterprise network meets the major design goals of segmentation, scalability, reliable inter-branch communication and fault tolerance. The six /28 departmental networks provide adequate addressing capacity for the 36 end devices, while /30 point-to-point links conserve address space. VLANs and router-on-a-stick provide logical separation, and OSPF enables dynamic routing across the redundant three-router backbone. Among the demonstrated physical/logical approaches, a hybrid enterprise architecture—star at the access layer with a redundant routed core—is the most suitable because it balances cost, scalability, manageability and reliability.

17. Submission Checklist

- ✓ 3 routers and 3 switches
- ✓ 36 PCs across three sites
- ✓ Six VLANs and six departmental subnets
- ✓ DHCP addressing plan
- ✓ OSPF routing plan
- ✓ Redundant triangular router backbone
- ✓ Star, mesh, bus and hybrid topology diagrams
- ✓ TCP/UDP/HTTP/DNS analysis
- ✓ Normal/high-traffic/congestion/failure test cases
- ✓ Latency, throughput, packet loss, jitter, delivery and recovery metrics
- ✓ Graphs and final evaluation
- ✓ User-provided Packet Tracer topology included